

Debt vs. Equity: Analysing the capital structure of Indian companies

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Abstract: Indian companies are known to have high rates of leverage and with the current crisis it is important to understand how companies, especially in financial stress raise capital. This paper provides a broad overview of capital structure of 2747 listed companies in India. For these firms the extent of solvency risk is estimated using Z score, which is found to be most accurate in the Indian context. The paper finds that companies with higher solvency risk tend to issue more corporate debt than bank borrowing and further equity market is dominated by the financial well performing companies. The paper examines in detail the factors that influence a company's choice of capital. It is observed that promoter control, size of leverage, corporate tax and reforms such as Insolvency and Bankruptcy Code have a significant bearing on a firm's choice of capital whereas solvency position of the position is not a significant consideration.

1. Identifying the problem

The current economic crisis has aggravated corporate financial stress. Despite the Reserve Bank's effort to quell distress through interest moratoriums many companies find themselves in need for additional capital to survive the downturn. The fact that nearly 39 per cent of the outstanding corporate loans payments opted for moratorium is a reflection of the degree of financial stress confronted by these firms² due to demand slowdown. To facilitate easy access to new capital, concurrent changes in regulations have been introduced by Securities Exchange Board of India (SEBI) for initial public offering, takeover and rights issues. These include changes to offer price and subscription conditions. Such regulatory changes are premised on importance of equity as a source of capital for revival. However, one may expect that this may not be the most accurate description of the securities market. Historically, there has been a preference among Indian companies to turn to intermediary based sources such as the bank (Jadhav, 2006). With the liberalisation of the securities market, there was an increase in number of issuances and value of capital raised until 1997(Jadhav, 2006). However, this slowed

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² <https://www.financialexpress.com/opinion/takeaways-from-rbis-financial-stability-report-on-moratorium/2036863/>

down and despite an increase in equity issuances and bank borrowing remains the largest sources of finance for companies till date. As will be shown in this study, corporate sector in India is significantly leveraged. However, even within the category of debt companies tend to rely on different kinds of instruments, which in turn can have varying implications for system wide impact. Capital structure of a firm is determined by considerations such as cost of capital, including tax implications, dilution of control and the regulatory compliance for capital issuance. While each firm uniquely factors in each consideration prior to capital issuance, this paper seeks to find major determinants of capital structure for companies in India. In addition this study seeks to find the impact of financial stress on decisions related to capital structure.

2. Decision to structure capital

A firm can source funds from the capital markets be it equity or debt, through borrowings from financial intermediaries or from internally generated surplus. The decision to use a particular structure of capital is based on a variety of considerations. Among the classic explanations for capital structure is that posited by Modigliani and Miller (1936). They radically transformed thinking on capital structuring by deriving the result that in an ideal world without taxes value of firm is independent of debt-equity mix. If debt and equity were treated on par for taxation the cash flows would not depend on structure. Thus the firm issuing capital in this scenario would choose either. However, interest payable on debt is tax deductible, although subject to limits in certain countries. Whereas equity in many countries is not deductible for tax purposes³. Therefore the choice of capital may depend on relative tax treatment of dividends, that can vary between classical system⁴, dividend relief system⁵ or imputation system⁶. Debt may gain preference in countries where classical system of dividend taxation exists. Since using more debt may be relatively tax efficient, it can help improve the bottom line. However, there are limits to which debt may be used to improve financial performance. High levels of debt exposure may make bank finance harder to access as well as exacerbate the insolvency risks

³ Except for Italy and Belgium where a deduction system was imagined.

⁴ Taxable at the corporate or individual level

⁵ Reduced rate at individual level

⁶ The recipient of dividend gets credit for the tax paid by corporate

associated with excessive leverage. Therefore, the capital mix depends on such consideration.

Then there is the maturity structure of capital which depends on supply side factors such as size of banking sector and maturity structure of banking liabilities. Usually short term liabilities can lead to higher uptake of short-term financing whereas countries with significant pension funds would take long-term finance. Companies also choose capital structure based on their preference to retain control, thus making debt relatively attractive where promoters are averse to dilution of control. This is an important consideration in emerging market economies, where family owned enterprises preferred to retain control (World Bank, 1994) or as is now observed in the case of Start-ups in India⁷. Another concern for companies is the periodic requirement to disclose information in cases where equity is raised through public offer and is listed on stock exchange. Then there are macroeconomic considerations- such as interest rates, inflation, government controls and state of domestic capital markets- that can influence the capital structure. Capital structure is not fixed and over time changes with the firm's stage of maturity. As some identify that firms seek to meet a target capital structure (Haas and Peeters, 2004). Myers (1984) proposed that the choice in capital structure follows an order, that is companies begin by using retained earnings then issue debt and finally issue shares. The focal point of this paper is to understand the structure of finance for companies in India. The choice is analysed by size of operations as well as financial performance. While above mentioned factors can have a bearing on the choice of capital structure, each firm chooses on the basis of its specific needs and strategy. Therefore, this paper identifies broad trends in capital structure over the last two decades. The impact of the factors such as taxation, control and more importantly degree of firm's financial stress on capital structure. For the analysis in this paper non-financial listed companies reporting information for at least 18 of the 20 years are selected. 2747 companies have been selected on this basis⁸.

2.1 Degree of Financial Leverage ratio in India

2.1.1 Deleveraging after IBC

In India companies have higher propensity to finance through borrowings. Fan, Titman and Twite (2012) estimate cross-country debt-equity ratio for firms to find Indian firms ranked sixth globally in leverage⁹ and on an average tended to borrow longer. In the

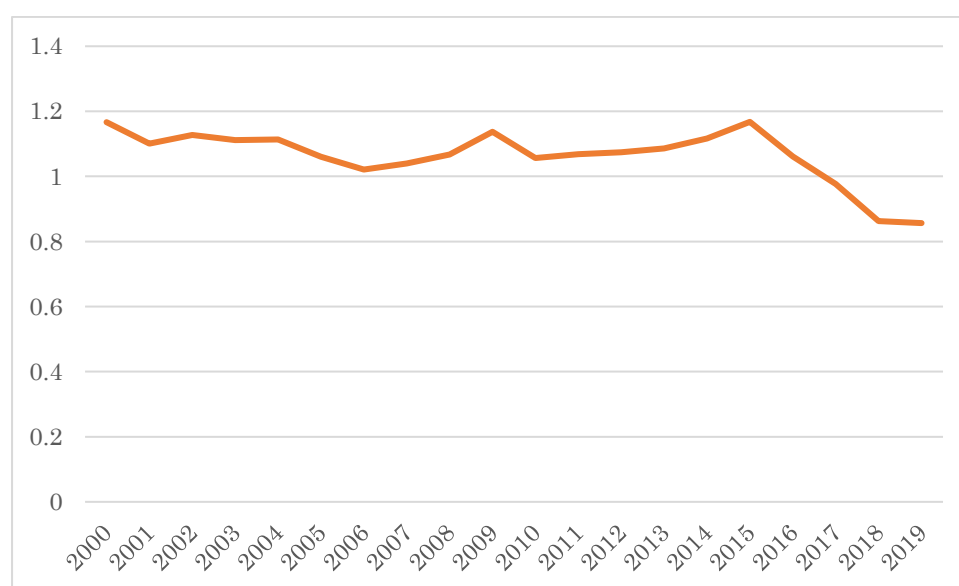
⁷ <https://www.livemint.com/companies/start-ups/indian-promoters-claw-back-control-11571761878443.html>

⁸ Check appendix for details

⁹ Page 33, <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.717.613&rep=rep1&type=pdf>

sample of companies selected for analysis in this paper, it is estimated that debt to equity ratio was greater than one¹⁰ between 2000 and 2016. Thus substantiating the claim that companies tend to be leveraged. However the share of short term borrowing in bank credit, at more than 60 per cent, is in fact substantial¹¹. In terms of trends it is seen that while the debt-equity ratio remained stable until 2015, a discernible shift was observed thereafter. Debt-equity ratio declined substantially across companies¹². The average debt to equity ratio declined from 1.16 in 2015 to 0.85 by March 2019.

Figure 1 Debt-equity ratio for Indian companies



Source: CMIE

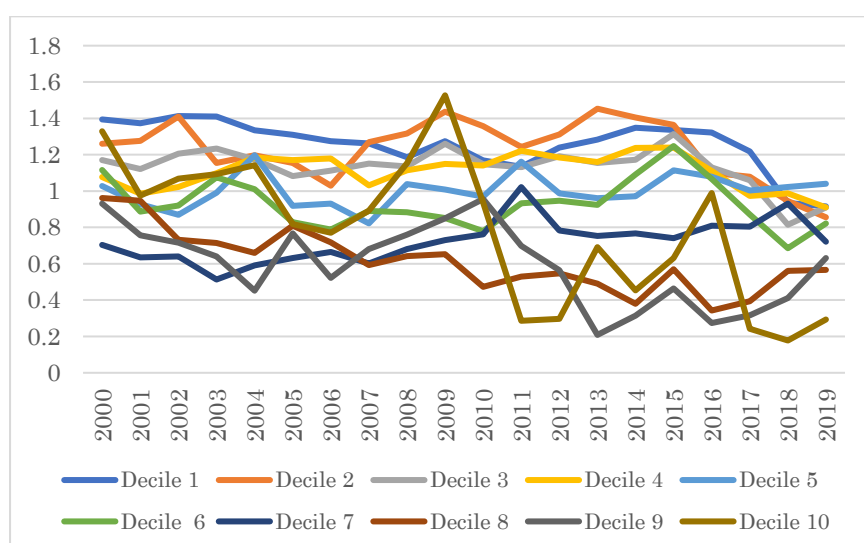
The degree of leverage also varies across firm size- measured in assets and sales. Companies can be categorised into deciles as per the size of turnover and assets. Using this classification provided by CMIE it is seen that during the past two decades, relatively large companies reported higher leverage ratio. That is firms within the first four deciles reported debt to equity of 1 or more which was significantly higher than the smaller companies. Throughout the period of analysis there remained a gap between the leverage of companies belonging to top and bottom three deciles. In line with the overall trend, deleveraging is also observed for top tier companies however companies belonging to decile 8-10 witnessed steady or even slightly increasing levels of leverage after 2016. Therefore the trend within the Indian industry was not uniform across size of companies.

¹⁰ Please check Appendix for description of dataset

¹¹ See Appendix point E

¹² This result was also observed for the full sample of companies.

Figure 2 Debt-equity ratio for Indian companies, decile wise



Source: CMIE

One may conjecture what may have culminated in the sharp reduction in proportions of debt used by companies. A possible explanation for this trend may be introduction of Insolvency and Bankruptcy Code in 2016¹³. The code put in place a system where firms with unsustainable levels of debt confronted the potential risk of insolvency. The risk that creditors now had swift legal recourse to settle defaults may have encouraged companies to deleverage. To confirm whether the shift is statistically significant, a t-test is conducted to compare the average leverage pre and post 2016. It is seen that the observed deleveraging is statistically significant at 2016 (Appendix Table A3). Yet, as is seen the lower deciles continued to take on more debt, this may perhaps have had to do with their lower debt-equity ratio.

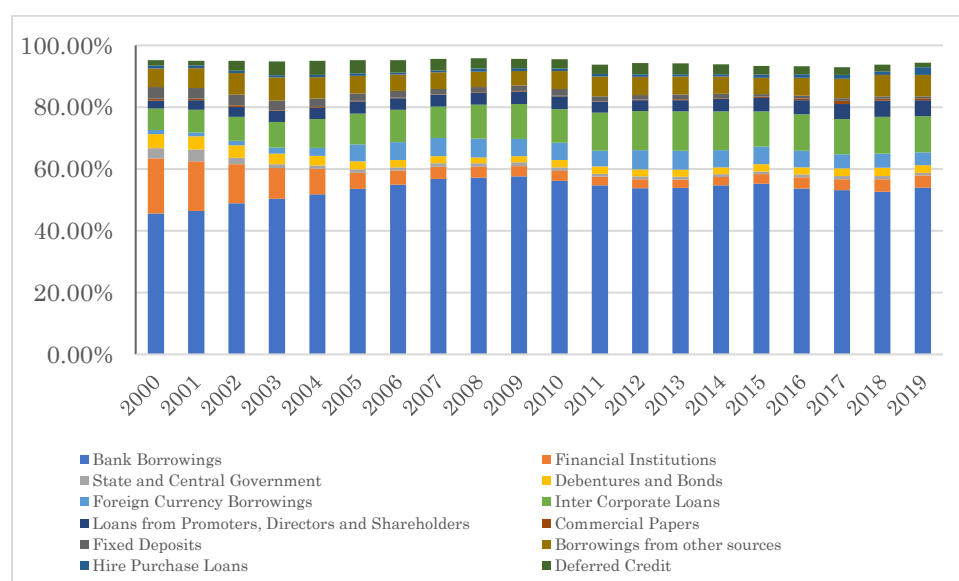
2.1.2 Structure of corporate debt

Despite the deleveraging, debt remains an important component of capital structure. For debt companies tend to rely on domestic banks, foreign currency loans or inter-corporate loans. This can depend on the interest rates, group structure, alternate investment opportunities as well as tax treatment of returns. It is observed that bank borrowing remains an important or dominant source of funding for Indian corporates. More than 50 per cent of the debt is sourced from banks. Inter-corporate loans comprising of loans from group companies, subsidiaries and other corporates are the next significant source of debt capital. The share of inter-corporate loans expanded until 2006, and has since remained

¹³ Note that since the data is for financial year end 2016, some of the changes may have been anticipatory

over 10 per cent. Over the past two decades the composition of inter-corporate loans has also changed significantly. From being sourced predominantly from other corporates these loans are now intra-group. This could be the result of corporate re-organisation and the lack of alternative investment opportunities. Thirdly, the share of financial institutions, comprising of asset management companies and insurance companies, declined significantly from 17 per cent in 2000 to 4 per cent in 2019. The decline in the lending by FIs began soon after the banking sector reforms were introduced in 2004. Meanwhile, Corporate debt, at less than 5 per cent, remained a small portion of the overall debt.

Figure 3 Share of corporate debt(%)

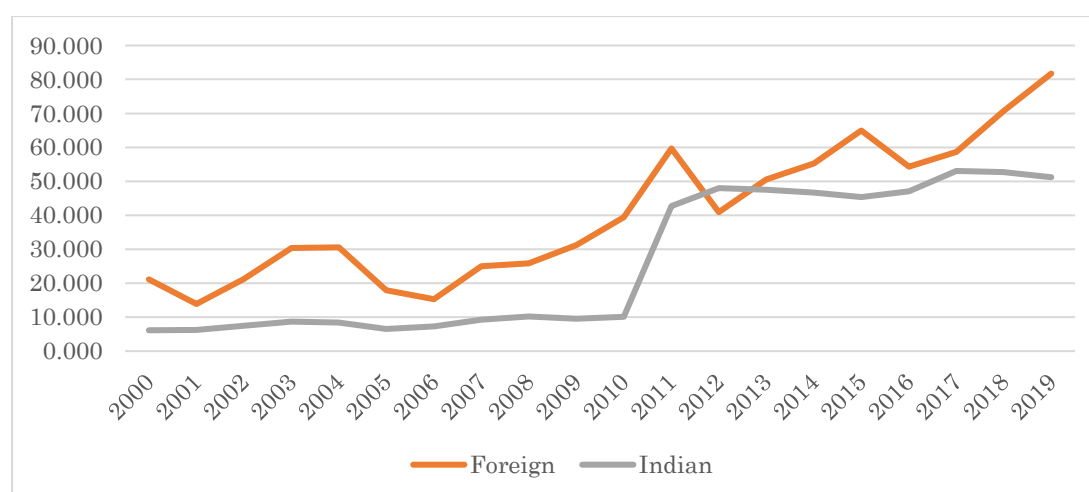


Source: CMIE

While the above graph presents an overall picture of the debt structure of all companies, it is important to mention that there are differences between the choice of capital between firms that are privately held, those with foreign ownership and those affiliated with a large business group. For example, foreign currency borrowing is an important source of debt for companies with foreign ownership. Inter-corporate loans, though equally large for companies with foreign ownership have been dominated by intra-group loans throughout the period. Moreover, Indian companies affiliated with large groups tend to behave similarly as foreign owned companies. For example, eighty large business groups in India are identified are those relying, in excess of 65 per cent, on inter-corporate loans for debt. The top ten among these are Reliance Group, HDFC Group, Bharti Telecom Group, LIC Group, Vedanta Group, Tata Group, Larsen and Toubro Group, Birla Aditya Group and ITC Group and Mahindra and Mahindra Group. In contrast other Indian

companies had demonstrably lower levels of intra-group loans. However, this changed significantly after 2010 after which Indian companies mimicked the features of foreign and group companies. These are not just the means to finance internally within a group structure, but can also be tax efficient, particularly when received from low tax foreign affiliates.

Figure 4 Share of Intra-Group loans in Inter-corporate loans 2000-2019



Source: CMIE

2.1.3 Retained earnings

Other than debt and equity, companies tend to rely on internal sources of finance i.e. retained earnings. That is a firm can choose to save a part of their profits and deploy these for expansion or to tide over financial strain. It is seen that on an average retained earnings formed a large proportion of assets for foreign companies, followed by unlisted companies. However, for all categories of companies the proportion of earnings retained increased significantly between 2003 and 2007, also a phase of high growth. However, there was a double dip in retained earnings one in 2008 and another in 2011, after which they declined. The sharp decline occurred during the global financial crisis and Europe's sovereign debt crisis. Interestingly, similar trends have been observed in other countries. For example, Alves (2018) shows that after 2008 companies located in 41 countries¹⁴ showed a declining trend in retained earnings. A decline in retained earnings can be for two reasons, worse financial performance as well as increased opportunities to invest. In

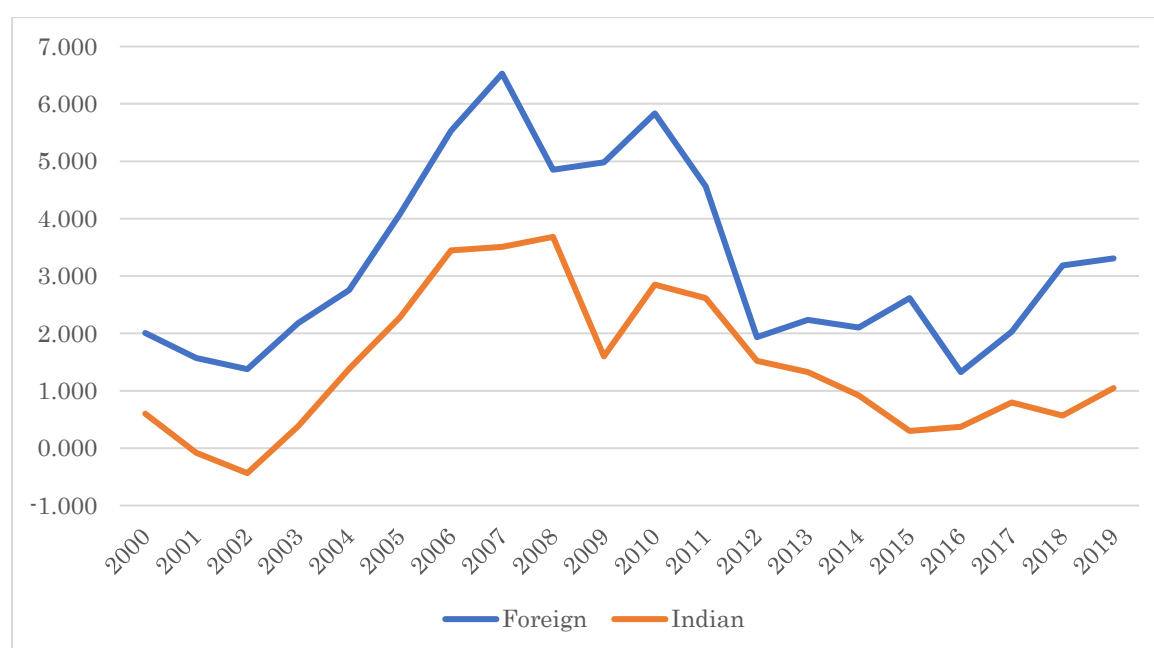
¹⁴ The sample consisted of 312,890 observations for 36,459 firms

During the years when retained earnings dipped, there was large spike in M&A purchases by India¹⁵. Thus change in trend can be attributed, in part, to an increase opportunity cost from retained profits, as observed at the end of the two crises. Interestingly the reversal in the decline began once companies were also seen deleveraging. Perhaps, the risk from leverage played an important role in the reversal of trend.

Retained earnings can also be used for purposed such as share buyback. can discourage retained earnings. At the margin, tax can also influence this decision. In 2013, the Finance Act introduced a buy-back tax¹⁶ at the rate of 20 per cent. In the following year the dividends paid out increased while retained earnings declined¹⁷.

Retained earnings remain an important source of funding for companies with foreign ownership (3% of total assets), as compared with Indian companies.

Figure 5 Average Retained Earnings to Total Assets (%)



Source: CMIE

¹⁵ See Appendix Figure A.

¹⁶ Section 115QA of the Income Tax Act 1961

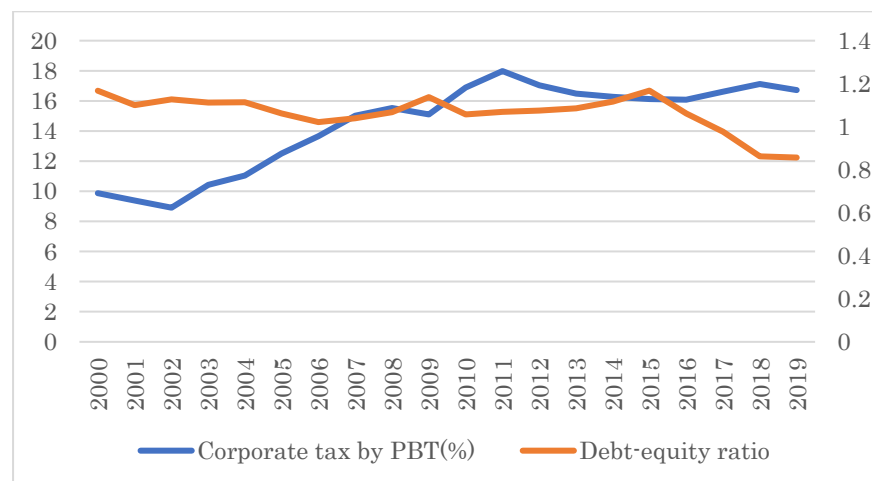
¹⁷ <https://www.law.ox.ac.uk/business-law-blog/blog/2020/05/taxation-income-capital-india-consistency-versus-priority>

3. Tax and capital structure

Based on the Modigliani and Miller's hypothesis, tax can influence the capital structure of a company. There is unending debate on whether debt's deductibility allows companies with higher rates of leverage to pay lower effective tax rates. This lop-sided treatment makes a case for equal treatment of debt and equity. However, the extent to which tax influences the decisions to finance may vary circumstantially. There are in fact other regulatory changes that can outweigh the tax advantage from such leverage. For example, the introduction of IBC caused such deleverage. However, given that dividends were until 2020 subject to a distribution tax, payable by the company, not available for credit against overall income tax. Whereas debt is deductible. Thus one may expect tax influences the choice of capita.

A simplistic representation is the trend in effective tax rates i.e. corporate tax to PBT and the debt equity ratio. Between 2000 and 2011, the effective tax rates paid by companies increased consistently. This was also a phase where the share of corporate tax in direct taxes increased. This rise in effective corporate tax rates was observed alongside lower debt -equity over the same period. The average effective tax rates for all listed companies rose to 17 per cent from 10 per cent. In the years after 2011, despite the deleverage, effective corporate tax rates were maintained at the higher levels achieved during the previous decade. As a result, after 2016 did not have a proportionate impact on effective tax rates. This could be the result of shift towards a gradual reduction in statutory corporate tax rates accompanied by phasing out of exemptions.

Figure 6 Corporate tax-PBT and Debt-equity ratio for companies



Source: CMIE

Thus there are phases when taxes are more closely associated with capital structure. Therefore to evaluate this the elasticity of tax rates to leverage is estimated for the select companies¹⁸. It is seen that while deleveraging by companies should result in higher effective corporate tax rates.

4. Financial stress and capital structure

A key feature of this paper is to examine sources of capital for companies in financial stress. This is important for two reasons-One it indicates markets that may signal swifter and better as economic slowdown. Two, this is indicative for policy that seeks to encourage wider availability of capital and the desirable regulatory changes. It is therefore imperative that financial stress be appropriately measured.

4.1 Measures of stress

In literature, the degree of financial stress in a company are measured using a variety of indices. Among the more popular measures, also conventional measure, is Altman's Z-score. This measure is a composite index¹⁹ of the working capital to total assets, retained earnings to total assets ratio, earnings before interest and tax to total assets, market value of equity to total liabilities and total sales to total assets ratio. All these indicators capture the solvency of the company. Therefore a company with a higher Z-score is less

¹⁸ See Appendix point B for results.

¹⁹ $Z = 1.2(\text{working capital to total assets}) + 1.4(\text{retained earnings to total assets ratio}) + 3.3(\text{earnings before interest and tax to total assets}) + 0.6(\text{market value of equity to total liabilities}) + 1.0(\text{total sales to total assets ratio})$

likely to face the risk of insolvency. Ideally, a score of more than 3 means the company is in the safe zone, whereas a company between 1.8 to 3 is considered to be in the grey zone and any score less than 1.8 represents acute distress. Over time more measures for financial distress evolved such as the Aggregate Bankruptcy Index (ABI) that modifies the Z score by ranking each of the indicators so as to make such index relative (Anderson, 2009). The ABI is between 0 and 1, where a higher score represents less likelihood of bankruptcy. The ABI is considered more robust since it follows a non-parametric approach, where no coefficient is estimated, thus avoiding complication such as multicollinearity between variables and avoids the restrictive normality assumption. However, since the focus of the paper is to estimate absolute levels of financial stress to identify firms that may be close to bankruptcy, Z-score is employed but as will be mentioned later its efficacy in predicting distress is evaluated using sample of companies referred for insolvency and bankruptcy. An important point is that these can be applied only to listed companies because there is no easily identifiable market price of an unlisted company.

There are other indices such as the Ohlson's O-score that predicts the likelihood of failure²⁰ using nine variables²¹- firm size, leverage, working capital, liquidity, profitability, change in net income, and debt financing. Then there is the Piotrotski's F-score that computes the financial score of companies. However, this is more to evaluate the firm from valuation perspective. There are nine indicators on which the company is assigned a value one if it meets the specified criterion and 0 otherwise. A financially sound company would score 9 whereas a value trap would score 0.

²⁰ $\exp(\text{O-score}) / 1 + \exp(\text{O-score})$

²¹ $T = -1.32 - 0.407 \log\left(T \frac{A_t}{GNP}\right) + 6.03 \frac{TL_t}{TA_t} - 1.43 \frac{WC_t}{TA_t} + 0.0757 \frac{CL_t}{CA_t} - 1.72 X - 2.37 \frac{NI_t}{TA_t} - 1.83 \frac{FFO_t}{TL_t} + 0.285Y - 0.521 \frac{NI_t - NI_{t-1}}{|NI_t| + |NI_{t-1}|}$

where

- TA = total assets
- GNP = gross national product price index level
- TL = total liabilities
- WC = working capital
- CL = current liabilities
- CA = current assets
- X = 1 if TL > TA, 0 otherwise
- NI = net income
- FFO = funds from operations
- Y = 1 if a net loss for the last two years, 0 otherwise

Table 1 Criterion for F-score

Variables	Criterion
Return on Assets	>0
Cash flow from operations	>0
Increase in return on assets	>0
Cash flow versus ROA	Cash flow > ROA
Long-term debt to assets	<0
Change in liquidity	>0
Stock issuance	<0
Gross margin	>0
Asset turnover	>0

There is a degree of overlap between the above mentioned, each with its limitations. It is therefore important to select a measure based on evidence in India.

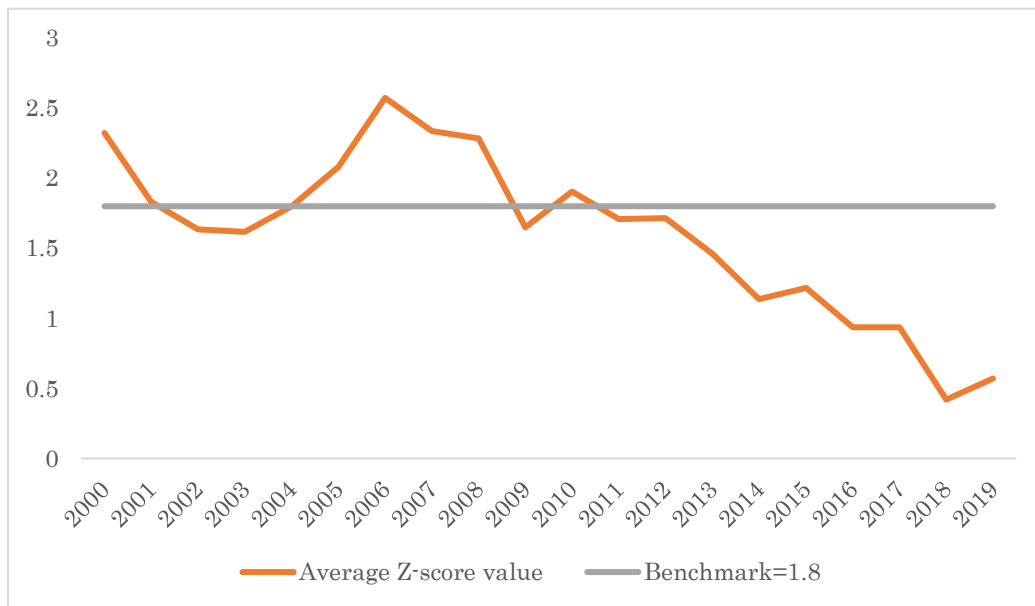
It is possible that a score can under-perform i.e. not pick up companies that have defaulted. This is Type I error, that is companies that should have been selected are excluded due to shortcomings of the model. On the flipside an index can over-state the likelihood of default, particularly since the estimated coefficients are imported from different contextual settings. This represents Type II error. That is, the variable is selected over-predicts the likelihood. To minimise these errors data on bankruptcy filings and annual reports on company performance are used. That is, the names of companies referred to in the list published Insolvency and bankruptcy Board of India's (IBBI) website are selected. Results from the indices are compared to assess which one of the index is able to identify the deterioration in financial performance of these companies²². An indicator that wrongly throws up well-performing companies is rejected. The O-score over predicts the likelihood of insolvency, that is it picks companies that have not defaulted. Therefore, for this reason O score is not used.

First, the Z-score is compared with F-score for 79 companies that were listed by IBBI. The average score for defaulting companies is calculated for companies and compared with the respective benchmark. It is seen from Figures 1 and 2 that the -score deteriorates much earlier than F score for the defaulting companies. This could be for two reasons, one that

²² Note that all such companies may not report information to Prowess CMIE therefore a sample of 100 companies that defaulted are taken. Then the indicator selecting maximum number of companies and at least a year prior to bankruptcy are selected.

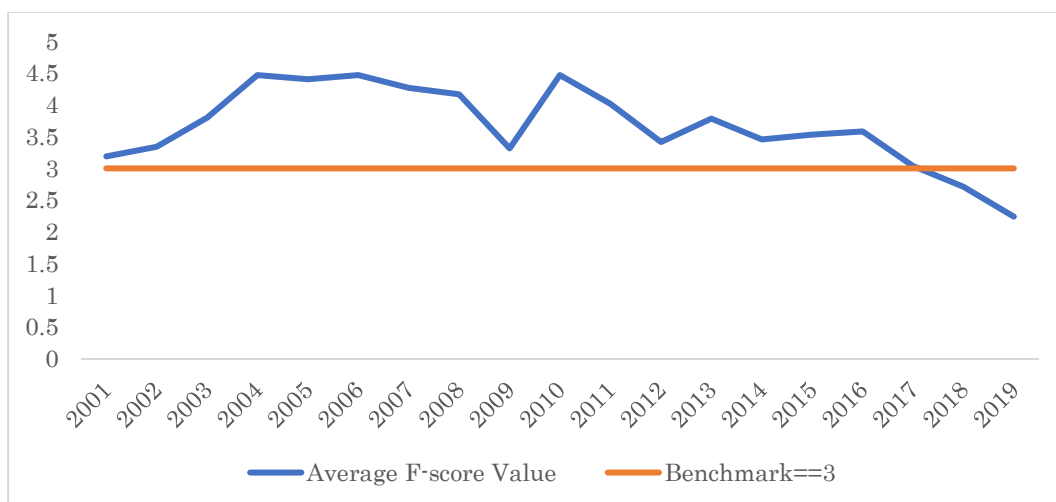
there is inclusion of companies that were legacy cases from the earlier regime, prone to delay. Two, this could mean that the liquidity of the firm deteriorates faster than the indicators for valuation. In fact it is estimated that on an average a company that defaulted in 2018, reported a deterioration in Z-score, below the threshold, 4.3 years prior to the default as against the 1 year for F-score.

Figure 7 Average Z score over the years for listed companies which defaulted after 2018



Source: Estimated from CMIE

Figure 8 Average F score over the years for Listed companies which defaulted after 2018



Source: Estimated from CMIE

An alternative method for evaluating these measures is to compare the number of firms that these measures identify among the list of defaulted companies. Taking 79 listed companies that matched, Z predicted more number of companies in most years except for in 2019.

Table 2 A Sample Of 79 Listed Companies Which Defaulted In Or After 2018

Year	Companies with Z SCORE<1.8 (1)	Companies with F SCORE<3 (2)	Share of (1) in 79 Companies	Share of (2) in 79 companies
2000	16		20.25%	
2001	23	29	29.11%	36.71%
2002	26	25	32.91%	31.65%
2003	28	19	35.44%	24.05%
2004	23	15	29.11%	18.99%
2005	21	17	26.58%	21.52%
2006	19	13	24.05%	16.46%
2007	28	12	35.44%	15.19%
2008	26	14	32.91%	17.72%
2009	37	26	46.84%	32.91%
2010	33	14	41.77%	17.72%
2011	43	13	54.43%	16.46%
2012	42	24	53.16%	30.38%
2013	50	26	63.29%	32.91%
2014	54	25	68.35%	31.65%
2015	53	22	67.09%	27.85%
2016	52	27	65.82%	34.18%
2017	55	34	69.62%	43.04%
2018	65	38	82.28%	48.10%
2019	46	49	58.23%	62.03%

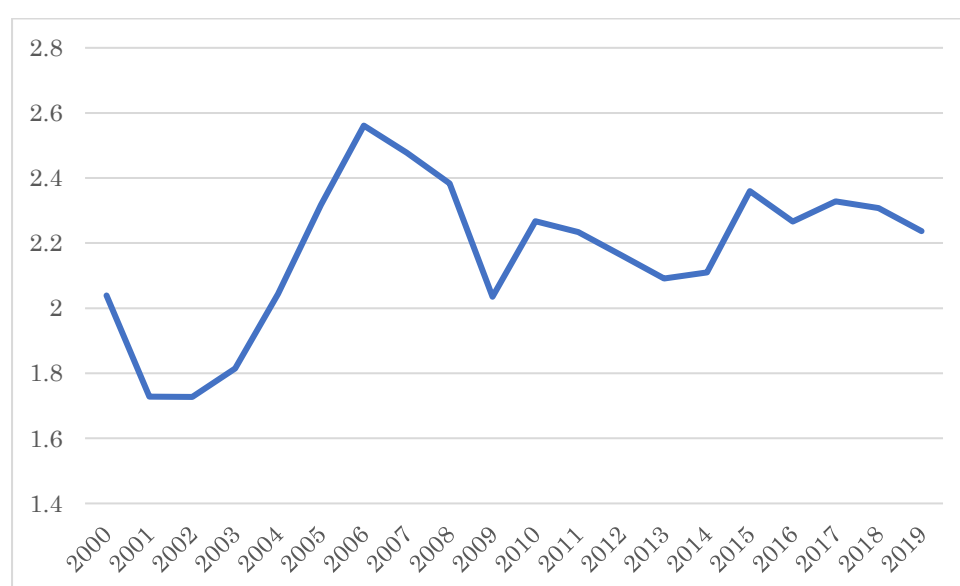
*The list of defaulted companies is matched with the dataset and then only listed companies are retained for comparison

The finding that Z score performs better than other measures of financial performance is corroborates earlier results such as Karamzadeh(2012) and Puro et al. (2013), that found that it performs better than O-score. Therefore in the analysis presented in this paper, Z score is selected.

4.2 Estimates of firm stress using Z-score

The average Z for listed companies improved in the years after 2004 and improved until 2006. With the financial crisis, Z-score of Indian companies deteriorated and a major decline was observed in 2009. Thereafter companies the score fluctuated until 2016, after which the score improved. This in part may have to do with the fact that the introduction of reforms such as the IBC may have compelled firms to take steps to improve the financial soundness of the business, as has already been shown in the case of leverage.

Figure 9 Average Z score of Listed Companies



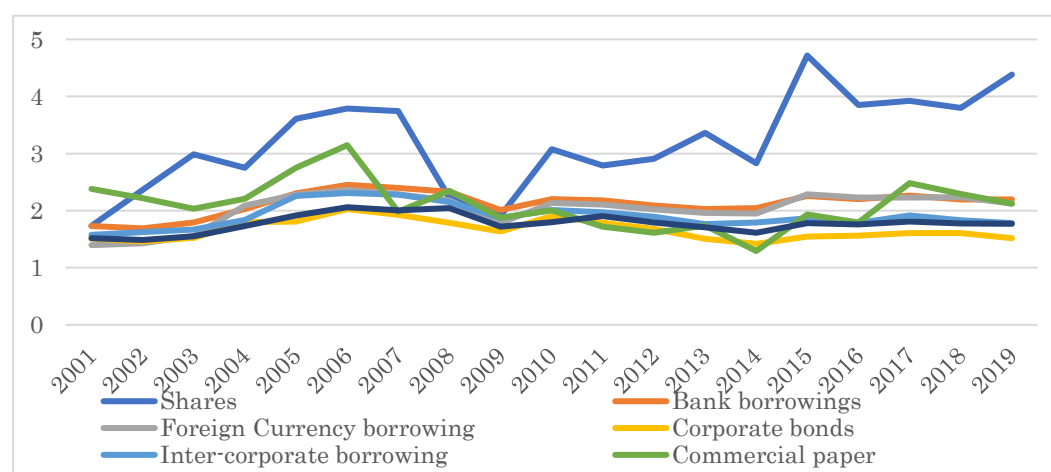
Source: CMIE

5. Capital choices by stress levels

A critical element of this paper is to find how stressed firms finance their activities. To a firm facing financial distress there are several alternatives available- company can be acquired or merge, or more equity can be raised either through the rights issue or a follow-on public offer. Then there is the alternative to raise money through additional bank borrowing, external commercial borrowing and corporate debt issuance. A company strapped for funding may pick one or many such sources. The choice as has been detailed earlier can be based on the tax treatment or regulatory framework. Therefore, taking the Z-score for each of the capital issuance it is seen

from Figure 10 that the average Z-score for companies issuing shares in any year is substantially higher than that for companies borrowing fresh capital from banks, in foreign currency or through corporate debt issuance. In fact, the corporate debt issuance fared the worst in terms of solvency. Whereas , commercial paper issuances after 2016 were by companies with better solvency. Nevertheless, in terms of total value of debt issuance, bank borrowing and corporate debt issuance remain large and incomparable with the other categories²³. In a hope to revive, companies can also choose to merge or may be acquired. In comparison to the companies that issued fresh capital-be it to stay solvent or expand operations- the companies that merged or were acquired had lower Z-score than bank and foreign currency borrowing until 2017(See Appendix).

Figure 10 Average Z-score for listed firms by type of capital issuance in each year



Source: Computed from CMIE

Thus a company may decide to raise capital for the purpose of raising fresh capital for expansion or for concerns of solvency. To meet each of these needs the type of issuance may vary. As has been shown graphically companies that rely on share capital tend to perform better on average than the others. However, capital issuance is not just determined by the financial stress but also promoter control and tax implications of debt versus that of equity. Therefore, in the next section, the likelihood of choosing one source of capital is predicted using promoters' shareholding, current debt equity, corporate tax rate and Z-score. Note that during the period of analysis statutory tax rate declines while effective tax rates increased, this could in part be the result of the phasing out of tax

²³ Fewer companies issued corporate debt and for that reason the average share in total borrowings is low whereas total value of capital raised is high.

exemptions. To proxy tax exemptions, fiscal benefits availed by a firm are taken to estimate the impact. The size of total assets are incorporated to examine if large companies tend to rely on particular source of capital. In addition, based on observed differences of deleverage post-IBC as well as exceptional reliance on inter-corporate loans among group companies, dummies are introduced. It is possible that the decision to raise capital is based on the observed financials in the previous year. Therefore two different specifications are estimated. One where the choice of capital is dependent on the variables in the same period and another where the choice is expressed as a lagged response. Since this study seeks to estimate the likelihood of choosing a particular form of capital, a panel logit model is estimated. Since the data consists largely of companies that issued only share capital are compared with those that purely issue debt. Another model is estimated with companies issuing only shares are compared with those issuing all other combinations.

$$P(\text{Share issuance} = 1|X_j)$$

1

$$= \frac{1}{e^{\beta_0 + \beta_1 \text{debt-equity ratio} + \beta_2 \text{share of promoters} + \beta_3 \text{corporate tax rate} + \beta_4 \text{fiscal benefits by corporate tax} + \beta_5 \text{z score} + \beta_6 \text{IBC dummy} + \beta_7 \text{group dummy}}}$$

Table 3 Prediction of likelihood of issuing fresh equity capital and borrowing

	Only Shares =1; Debt issuance in other combinations=0		Only Equity=1 Only debt=0 (for only pure issuance)	
Debt-equity ratio	-74.22*** (-4.05)		104*** (22.34)	
Debt-equity ratio (t-1)		-12.24*** (2.71)		-13.96*** (3.16)
Shares held by promoters	-0.022* (-1.85)		-0.03** (0.102)	
Shares held by promoters(t-1)		-0.0072 (0.012)		-0.016* (0.015)

Statutory corporate tax rate	-0.023 (-0.27)		-0.045* (-0.102)	
Statutory corporate tax rate(t-1)		-0.72*** (0.272)		-0.692** (0.28)
Z-score	0.158** (2.1)		0.11 (0.204)	
Z-score(t-1)		0.0157 (0.08)		0.135 (0.08)
Dummy for large business groups	0.14 (0.33)	0.177 (0.397)	0.099 (0.46)	0.183 (0.672)
Dummy for IBC	0.571* (1.66)	1.657*** (0.484)	0.74* (0.425)	1.35*** (0.504)
Log of total assets	0.334 (1.4)		-0.27 (0.26)	
Log of total assets (t-1)		0.393** (0.222)		0.486** (0.251)
Fiscal benefits by corporate tax	0.0013 (1.42)		0.0017 (0.002)	
Fiscal benefits by corporate tax (t-1)		0.00015 (0.0018)		0.0001 (0.0017)
Constant	-3.485 (-1.08)	18.77** (9.14)	-0.12 (3.83)	18.9* (9.7)

Note: * implies significant at 10%, ** implies significant at 5% and *** implies significant at 1%

The results reported in Table 3 show that the companies that an increase in the leverage of the firm reduced the odds of share issuance significantly. This could be since high levels of leverage may adversely affect the valuation of a company reducing it chances of sourcing capital from the equity market. The results also corroborate the hypothesis that companies with higher share of promoter control may have a lower likelihood of issuing share capital in order to retain control. Earlier in this paper it was shown that companies deleveraged after IBC. The dummy for years after IBC's implementation shows that the

odds of share of capital issuance did increase. As for the corporate tax rate, a company is likelier to take on more debt in a year based on previous year's tax rate however the fiscal benefits do not impact the outcome significantly. It can also be inferred that large firms, measured by the size of their assets are also likelier to leverage, however this difference is not apparent between large group firms and others. A variable of key interest was the z-score. Only in one specification it is found that better performing firms have higher odds of issuing shares. This means that evidence does not support the hypothesis that company takes consideration its solvency while making the decision to issue a particular kind of capital.

6. Conclusion

In India debt remains an important source of funding for firms and as is evident companies with higher debt-equity ratio are more likely borrow more. It is paradoxical that companies that were already leveraged were likelier to resort to borrowing. Further, it is observed the composition of debt also varied across firm type, i.e. inter-corporate loans were higher among group companies and foreign companies.

There can be various factors at play that could influence the choice of capital, among these this paper explores promoter's control, reforms such as IBC and tax. In addition the paper also seeks to ask where are companies with solvency risks likely to go to raise capital.

Interestingly, preliminary statistical findings suggest that in any year a company facing acute financial stressed resorts to issuing corporate bonds. Whereas companies with outstanding solvency condition tend to issue equity. This points to the exclusivity of the equity markets to well-performing companies. However, when considering the problem from the company's point of view, i.e. what capital structure it would choose given the various consideration it is observed that solvency is insignificant in comparison to factors such as current or past leverage, corporate tax rate, promoter control and reforms such as the IBC. An interesting finding of this paper is that the IBC may have resulted in a behavioural shift in how the company raises capital.

Between 2000 and 2019 more capital was raised through private and qualified placements than public issues. During this period, there were 1484 public issues as compared to 48,895 private and qualified institutional placement. The amounts raised were also respectively higher, the amount raised through the latter were INR 69,115 billion whereas for the former this amount was INR 8610 billion. This would suggest that

the regulatory requirements including disclosures and time taken discourage firms from resorting to public offer are also important. However, such analysis is not possible has not been considered.

Appendix

A. Prowess IQ reports statistics on 51062 companies. However, for the analysis in this paper, 4112 companies are selected. Since the paper focusses largely on measuring financial distress and to analyse capital structure decisions for these companies, 4112 companies are selected. The method of selecting companies is that the information is complete for at least 18 out of 20 years. Financial firms have been excluded from the dataset since these would have capital requirements that are different from the manufacturing and services companies. This is to ensure that the comparison is for the same set companies across years. Then our sample is restricted to listed companies leaving us with 2747 companies.

B. Panel regression results, using random effects, for corporate tax-PBT (%) and Debt to Equity(%). Note than Hausman test is used to determine that random effects model is more appropriate. It is theoretically sound to employ random effects since firms are affected by general economic conditions as well as there are firm specific effects.

Table A2. Regression results*

	Listed Companies
N	2747
R ²	0.02
Explanatory variables	Corporate tax to PBT(%)
Debt to Equity(%)	-1.32*** (0.046)
Alpha	17.41*** (0.192)

*Value in error, at 1%.

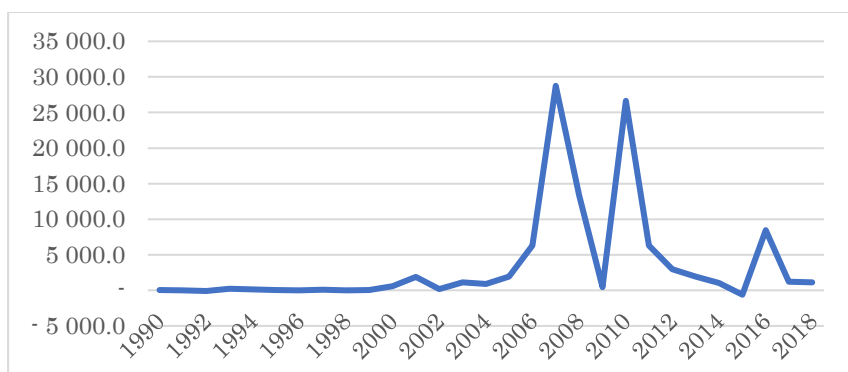
brackets represent standard three stars imply significant

C. We divide the dataset into two periods, one between 2000-2015 and one after 2015. For the firms it is estimated if the difference in leverage is statistically different.

Table A3 t-test for difference between average debt-equity

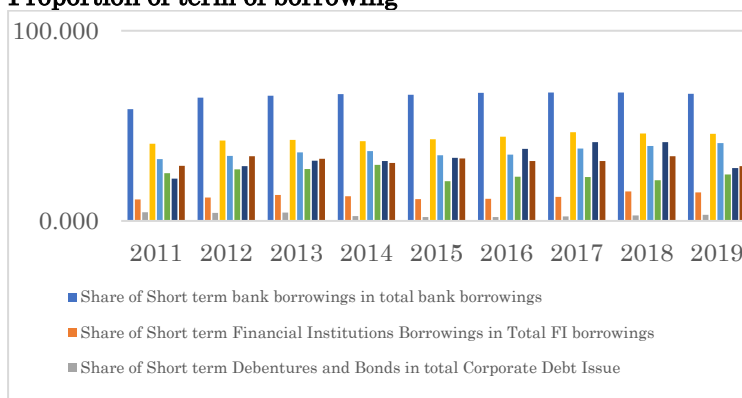
Period	Average debt- equity	N	Standard error
2000 to 2015	1.09	37128	0.009
After 2015	0.941	8843	0.02
t- value	7.126		

D. M&A purchases by India (Million US Dollars)



Source: UNCTAD

E. Proportion of term of borrowing



Source: Computed from CMIE

F. Average Value of Z-score of Acquired Companies up to the year of Merging

Year of Merging					
	2015	2016	2017	2018	2019
2001	1.652	1.778	1.892	4.973	1.877
2002	2.234	1.841	1.702	3.714	2.005
2003	2.123	1.976	1.905	2.704	2.329
2004	2.106	2.544	2.242	2.521	2.370
2005	2.642	2.883	3.571	2.834	2.618
2006	1.806	2.769	3.071	2.891	2.703
2007	2.333	2.325	3.165	2.466	2.633
2008	2.122	2.004	2.513	2.178	2.345
2009	1.836	1.708	1.831	1.993	3.505
2010	2.330	1.499	2.019	2.058	2.570

2011	2.177	1.554	1.746	2.301	2.595
2012	2.163	1.478	1.272	2.225	2.398
2013	2.244	1.538	1.194	2.355	2.562
2014	2.215	1.627	1.122	2.334	3.148
2015		1.476	1.324	2.757	4.048
2016			1.652	2.090	3.533
2017				2.818	2.965
2018					3.036
2019					
2020					

G. Average Z-score of companies that were acquired

